

Amazon EC2OverviewFeaturesPricingInstance TypesFAQsGetting StartedResources

Amazon EC2

Secure and resizable compute capacity to support virtually any workload

Get started with Amazon EC2

Over 400 instances for virtually every business need

Only cloud provider that supports macOS

25 regions and 81 availability zones globally

Choice of Intel, AMD, and Arm-based processors

New AWS Local Zones in Atlanta, New York City, and Portland | Now Generally Available in 13 cities across the United States »

Reliable, scalable, infrastructure on demand

- Increase or decrease capacity within minutes, not hours or days
- SLA commitment of 99.99% availability for each Amazon EC2 region. Each region consists of at least 3 availability zones
- The AWS Region/AZ model has been recognized by Gartner as the recommended approach for running enterprise applications that require high availability



"Since March, we have seen market volumes increase by 2-3x...We automatically turn on and off up to 100,000 compute nodes in a single day. We would have been unable to handle this surge in volume within our on-premises data center."

Steve Randich
Executive Vice President and Chief Information Officer, FINRA

Secure compute for your applications

- With the [AWS Nitro System](#) virtualization resources are offloaded to dedicated hardware and software minimizing the attack surface
- Lockdown security model prohibits administrative access eliminating possibility of human error and tampering
- AWS supports 89 security standards and compliance certifications including PCI-DSS, HIPAA/HITECH, FedRAMP, GDPR, FIPS 140-2, and NIST 800-171, which is meaningfully more than any other cloud provider



"We looked at many public clouds but AWS scored most highly against our key criteria, which included stability, security, and breadth of services."

Lawrence Fong
GM IT Solutions, Cathay Pacific

Flexible options to optimize cost

- With On Demand, pay for only the compute you need. Gain up to 40% better price performance with [AWS Graviton2-based instances](#) or 10% lower costs with [AMD-based instances](#).
- [Amazon EC2 Spot](#) can reduce costs up to 90% or accelerate performance for fault-tolerant workloads such as big data, containers, web services, and CI/CD.
- [AWS Savings Plans](#) offers savings of up to 72% on Amazon EC2 instances usage, regardless of instance family, size, OS, tenancy or AWS Region.



"We tested the new M6g instances using industry standard Lmbench and certain Java benchmarks and saw up to 50% improvement over M5 instances. We're excited about the introduction of AWS Graviton2-based Amazon EC2 instances."

Ed Hunter
Director of Performance and Operating Systems, Netflix

Easy to migrate and build apps

- Get started quickly through [AWS Migration Tools](#), [AWS Managed Services](#), or [Amazon Lightsail](#)
- [AWS Professional Services](#), [AWS Support](#), and [AWS Training and Certification](#) help you along the way
- Tens of thousands of [APN Partners](#) to help companies build, market, and sell their AWS offerings



"Management was easy—it was very simple to upload a machine to the platform and simply adjust sizing according to need afterwards."

Sadao Matsuda
SAP IT Manager, VWCO

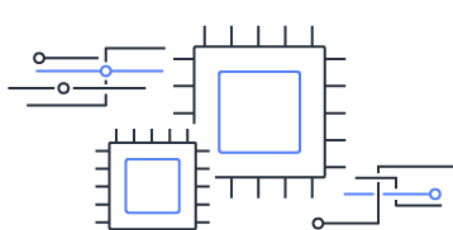
Building Blocks

Amazon EC2 offers the broadest and deepest choice of instances, built on the latest compute, storage, and networking technologies and engineered for high performance and security.



Faster innovation and increased security with AWS Nitro System

The [AWS Nitro System](#) is the underlying platform for our next generation of EC2 instances that offloads many of the traditional virtualization functions to dedicated hardware and software to deliver high performance, high availability, and high security while also reducing virtualization overhead. The Nitro System is a rich collection of building blocks that can be assembled in many different ways, giving us the flexibility to design and rapidly deliver new EC2 instance types with an ever-broadening selection of compute, storage, memory, and networking options.



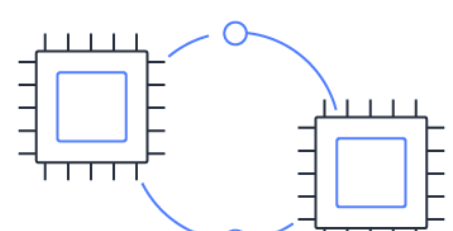
Choice of processors

A choice of latest generation [Intel Xeon](#), [AMD EPYC](#), and [AWS Graviton](#) CPUs enables you to find the best balance of performance and price for your workloads. EC2 instances powered by [NVIDIA GPUs](#) and [AWS Inferentia](#) are also available for workloads that require accelerated computing such as machine learning, gaming, and graphic intensive applications.



High performance storage

[Amazon Elastic Block Store](#) (EBS) provides easy to use, high performance block storage for use with Amazon EC2. Amazon EBS is available in a range of volume types that allow you to optimize storage performance and cost for your workloads. Many EC2 instance types also come with options for local NVMe SSD storage for applications that require low latency.



Enhanced networking

AWS is the first and only cloud to offer 400 Gbps enhanced Ethernet networking for compute instances. [Enhanced networking](#) enables you to get significantly higher packet per second (PPS), lower network jitter, and lower latency. For high performance computing (HPC) applications, Elastic Fabric Adapter is a network interface for Amazon EC2 instances that offers low-latency, high-bandwidth interconnect between compute nodes to help scale applications to thousands of cores.



Choice of purchasing model

We offer a choice of multiple purchasing models with On-Demand, [Spot Instances](#), and [Savings Plan](#). With Spot Instances, you can save up to 90% for fault-tolerant workloads. With Savings Plan, you can save up to 72% savings with committed usage and flexibility across EC2, Fargate, and Lambda. You can also optimize your costs with recommendations on instances, built into EC2 with [AWS Compute Optimizer](#) or through tools such as [Cost Explorer](#).

Case Studies



What's New

[Amazon EC2 Fleet and Spot Fleet now support automatic instance termination with Capacity Rebalancing](#)

11/05/2021

[Amazon EC2 now supports access to Red Hat Knowledgebase](#)

11/03/2021

[AWS Graviton2 based T4g instances are now available in AWS GovCloud \(US-West\) Region](#)

11/02/2021

[Amazon EC2 now supports sharing Amazon Machine Images across AWS Organizations and Organizational Units](#)

10/29/2021

[Introducing Amazon EC2 C6i instances](#)

10/28/2021

[Amazon EC2 R5b instances are now available in 2 additional regions](#)

10/28/2021

[Amazon EC2 announces attribute-based instance type selection for Auto Scaling groups, EC2 Fleet, and Spot Fleet](#)

10/27/2021

[Introducing Amazon EC2 Spot placement score](#)

10/27/2021

[Announcing the General Availability of AWS Local Zones in Las Vegas, New York City, and Portland](#)

10/26/2021

[Announcing Amazon EC2 DL1 instances for cost efficient training of deep learning models](#)

10/26/2021

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Check out the EC2 product features

Learn more about features for instance security, storage, networking, autoscaling, and more

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Get started with EC2

Follow these steps to get started with Amazon EC2.

[Learn more »](#)



Dive deep into EC2 resources and tools

Learn tips and techniques to get the most out of your experience with Amazon EC2.

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